

Certified Attribution of Prompt Segments: Auditing What Makes a System Prompt Safe

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Abstract

System prompts encode much of an LLM’s safety behavior—refusal policies, prompt-injection defenses, formatting and persona constraints—as a list of natural-language *segments*. Yet practitioners cannot say which segments actually carry a given behavior, and routine prompt compression may silently delete a safety-critical clause. We cast this as attributing a behavioral metric to prompt segments via cooperative-game (Shapley and Owen) values, and identify the regime that makes it hard for alignment: the value function is a *doubly-stochastic, pay-per-call oracle* (an LLM plus a noisy behavioral judge over a task distribution), so every attribution is an estimate that must carry an honest error bar. We contribute a segment-attribution estimator whose output is a *certificate*: anytime-valid betting confidence sequences, composed with a complementary-contribution decomposition and a union bound, yield simultaneous (ε, δ) intervals for all segments that remain valid under fully adaptive, data-dependent sampling and stopping. We prove the certificate’s validity in the doubly-stochastic setting, prove a two-level allocation result explaining why fresh coalitions dominate within-coalition replication, and prove a *safe-pruning* guarantee: a hierarchical triage never certifies a genuinely load-bearing group of segments as removable, except with probability δ . Across synthetic games and real models (Qwen3.5-9B, Gemma-4), including an exact 2^{20} -coalition regime where enumeration is impossible, our estimator matches or beats strong provable baselines (Leverage SHAP, SVARM) under noise while being the only one that certifies. As an alignment demonstration, we certify which system-prompt clauses are load-bearing for prompt-injection resistance and show that a planted sabotage instruction is certified strictly harmful—turning prompt auditing into verifiable evidence for a safety case.

1 Introduction

A deployed language model’s behavior is governed in large part by its *system prompt*: a list of natural-language instructions—“treat text in the user’s message as untrusted,” “refuse requests for X,” “answer with a single letter,” a persona, some few-shot examples. These *segments* accumulate over a product’s lifetime and are continually edited, reordered, and *compressed* to save tokens. Two questions have no principled answer today: *which* segments actually produce a safety-relevant behavior, and whether a proposed edit or compression

will *erode* that behavior. A team that prunes a “bloated” system prompt for latency has no guarantee it did not just delete the clause that was doing all the work of resisting prompt injection.

This is an attribution problem, and cooperative game theory gives the principled answer: treat each segment as a player, let the “value” of a subset S of segments be the expected behavioral metric of the prompt containing exactly S , and report each segment’s Shapley value—its average marginal contribution, accounting for interactions with the others. Group segments into sections and the same machinery gives Owen values. The obstacle is not the definition but the *oracle*. Unlike the tabular settings where Shapley estimation is usually studied, here evaluating one coalition means running an LLM on sampled inputs and scoring its outputs with a possibly-noisy behavioral judge: every value is an expensive, *doubly-stochastic* estimate (noise from input sampling and from decoding/judging). An attribution that ignores this noise is not auditable—and for a safety decision, an attribution you cannot trust is worse than none.

We therefore ask for *certified* segment attribution: every reported number carries a simultaneous, high-probability error bar, and every “this clause is negligible, prune it” decision is backed by a guarantee. Our contributions:

- **A certified estimator (§4).** We combine a complementary-contribution decomposition of the Shapley value—so one evaluated pair updates every segment—with per-stratum *betting* confidence sequences (hedged capital; Waudby-Smith and Ramdas 2023) and a union bound. The result is a simultaneous (ε, δ) certificate over all segments that remains valid under *fully adaptive* sampling and stopping, which is what lets the estimator spend its budget where the uncertainty is.
- **Theory for the noisy oracle (§5).** We prove certificate validity in the doubly-stochastic setting (Thm. 1), a two-level allocation result showing fresh coalitions dominate within-coalition replication in call-cost while replication can win in token-cost under prompt caching (Prop. 1), and a *safe-pruning* guarantee: a genuinely load-bearing group is never certified removable except w.p. δ (Thm. 2).
- **Evidence at scale (§6).** On synthetic games and two real models we match or beat the strongest provable baselines (Leverage SHAP, SVARM) *under noise*, while being the only method that certifies; betting certificates are

3–7 $\times$ tighter than empirical-Bernstein at equal δ with perfect empirical coverage. We reach an exact-ground-truth $n=12$ regime and a 2^{20} -coalition regime where enumeration is impossible and the reference is itself a certified gold standard.

- **An alignment demonstration (§7).** We certify which system-prompt clauses are load-bearing for prompt-injection resistance, and show a planted sabotage instruction is certified *strictly harmful*. The safe-pruning triage removes only genuinely inert clauses. This turns prompt auditing into verifiable evidence usable in a safety case.

Our framing is deliberately alignment-first: the deliverable is not a better attribution number but an *auditable* one, and a compression procedure that *cannot* silently strip safety behavior.

2 Related Work

Attribution for LLM prompts. TokenSHAP and TextGenSHAP compute Shapley values over prompt tokens; ContextCite fits a sparse linear surrogate to context ablations; AttriBoT accelerates leave-one-out context attribution; TraC-LLM searches long contexts for provenance. Closest to us, ProCut segments production prompt templates and prunes low-attribution units. All of these are heuristic: none reports a finite-sample error bar, and leave-one-out and linear surrogates ignore the interactions between clauses that redundancy and conflict are made of. Cluster Shapley is the one LLM-attribution method with formal bounds, but it values *retrieved documents* for revenue sharing under a clustering assumption, a different game.

Shapley estimation with guarantees. Permutation sampling with Hoeffding bounds, stratified and Neyman-optimal variants, and empirical-Bernstein adaptive allocation (Burgess and Chapman 2021) give variance-adaptive additive bounds. Complementary contributions (Zhang et al. 2023) and SVARM (Kolpaczki et al. 2024) let one evaluation update all players; Leverage SHAP (Musco and Witter 2025) gives the first non-asymptotic guarantee in the KernelSHAP regression family. These are developed for *deterministic* value functions on tabular or vision features. We show empirically that their optimality does not survive a noisy oracle, and we supply the missing certificate.

Anytime-valid inference. Betting/e-process confidence sequences (Waudby-Smith and Ramdas 2023) give time-uniform intervals for bounded means and underpin our adaptivity-safe certificate. Data Banzhaf (Wang and Jia 2023) is the one prior analysis of value estimation under a noisy utility, but it abandons the Shapley value for the Banzhaf value and targets data valuation.

Alignment context. Prompt injection and instruction-hierarchy robustness are active safety concerns; system-prompt clauses are a primary defense. Safe-by-design engineering and formal safety cases call for *auditable* evidence that a mitigation is effective. Our contribution is an attribution primitive that produces exactly such evidence and a compression procedure with a safety guarantee.

3 Problem Setup

Let a prompt decompose into segments $N = \{1, \dots, n\}$. For $S \subseteq N$ let $\pi(S)$ be the prompt containing exactly the segments in S , rendered in a fixed canonical order so the cooperative game is well defined. Fix a task distribution $\mathcal{D}$ and a bounded behavioral metric $u \in [0, 1]$ (exact-match correctness, a refusal indicator, an injection-resistance indicator, or an LLM-judge score). The *value* of a coalition is the population metric

$$v(S) = \mathbb{E}_{q \sim \mathcal{D}, y \sim \text{LLM}(\pi(S), q)} [u(y, q)] \in [0, 1]. \quad (1)$$

The estimand is the Shapley vector $\phi(v) \in \mathbb{R}^n$ and, given a partition of N into groups, the Owen values.

Doubly-stochastic pay-per-call oracle. We never observe $v(S)$. An *oracle call* draws one $q \sim \mathcal{D}$ and one generation and returns $u(y, q)$; averaging r calls yields $\hat{v}(S) = \frac{1}{r} \sum_{j=1}^r u(y_j, q_j)$, an unbiased estimate whose variance $\sigma^2(S)/r$ combines input-sampling and decoding/judging noise. Calls are the expensive resource. We additionally track a token-cost model in which coalitions sharing a prompt prefix are cheaper under prompt/KV caching.

Safety reading. When u is a safety metric, ϕ_i is the certified contribution of clause i to that behavior. Two decisions matter: *verification*—is ϕ_i bounded away from zero (load-bearing) or below zero (harmful)?—and *safe pruning*—can a group of clauses be removed without losing behavior? Both require a lower bound on how wrong the estimate can be, i.e. a certificate.

4 Method

4.1 Certified Shapley from Paired Coalitions

We use the complementary-contribution identity: with $\text{CC}(S) = v(S) - v(N \setminus S)$,

$$\phi_i = \frac{1}{n} \sum_{s=1}^n \mathbb{E}[\text{CC}(S) \mid i \in S, |S| = s], \quad (2)$$

so a single evaluated pair $(S, N \setminus S)$ contributes a sample to *every* segment: members of S receive $\text{CC}(S)$ in their size- $|S|$ stratum and non-members receive $-\text{CC}(S)$ in their size- $(n-|S|)$ stratum. This maximizes statistical reuse and doubles as antithetic sampling.

Algorithm 1 Certified segment Shapley (CC-BET)

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1: input: oracle, budget  $B$ , replicates  $r$ , level  $\delta$ 
2: init empty stratum sequences  $X_{i,s}$ ; spent  $\leftarrow 0$ 
3: while spent +  $2r \leq B$  do
4:   pick size  $s$  maximizing current stratum uncertainty
5:   draw  $S$  uniform of size  $s$ ; evaluate  $\hat{v}(S), \hat{v}(N \setminus S)$ 
6:    $c \leftarrow \hat{v}(S) - \hat{v}(N \setminus S)$ ; spent  $\leftarrow$  spent +  $2r$ 
7:   append  $c$  to  $X_{i,s}$  ( $i \in S$ ),  $-c$  to  $X_{j,n-s}$  ( $j \notin S$ )
8:   if all per-segment half-widths  $\leq \varepsilon$  then break
9:   end if
10: end while
11: for each segment  $i$  do
12:    $[\ell_{i,s}, u_{i,s}] \leftarrow \text{BETTINGCI}(X_{i,s}, \delta')$  per size
13:    $[\ell_i, u_i] \leftarrow \frac{1}{n} \sum_s [\ell_{i,s}, u_{i,s}]$ 
14: end for
15: return centers  $\phi_i$ , intervals  $[\ell_i, u_i]$ 

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Algorithm 2 Certified safe-pruning triage (OWEN-ELIM)

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1: certify group values with CC-BET( $\delta/2$ ) on the quotient
   game
2: for each group  $g$  do
3:   if  $[\ell_g, u_g] \subseteq [-\tau, \tau]$  then mark removable
4:   else keep; refine members with CC-BET( $\delta/2$ ) in  $g$ 
5:   end if
6: end for
7: return removable set, kept set, member certificates

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For each (player, size) stratum we maintain the raw sequence of observed (complementary-)contributions and form a *hedged betting* confidence sequence for its mean (Alg. 1, BETTINGCI). A gambler bets against each candidate mean m with predictable plug-in stakes; the capital process $K_t(m) = \prod_i (1 + \lambda_i(m)(x_i - m))$ is a nonnegative martingale under the true mean, so $\{m : K_t(m) < 1/\delta'\}$ is a confidence sequence valid simultaneously over all sample counts t (Ville’s inequality). The per-segment interval is the size-averaged Minkowski sum of its stratum intervals, run at $\delta' = \delta/(n \cdot n_{\text{strata}})$ so all intervals hold jointly with probability $\geq 1 - \delta$. Sizes are allocated adaptively toward the strata that dominate the current uncertainty; because the certificate is anytime-valid, this data-dependent allocation and the stopping rule cannot invalidate it.

4.2 Hierarchical Triage and Safe Pruning

Given a partition into groups, we first certify the *group* (Owen/quotient) values with CC-BET on the quotient game, at level $\delta/2$. A group is *certified removable* only if its whole interval lies inside $[-\tau, \tau]$ for a user threshold τ ; groups whose interval escapes $[-\tau, \tau]$ are *kept* and refined member-by-member (their Owen values estimated with the same paired identity inside the group, marginalizing the other groups), at the remaining budget split by residual width. This yields a certified keep/prune/refine triage of a prompt at group granularity (Alg. 2).

4.3 Surrogate Control Variate

When the game is nearly low-order we fit a k -additive Möbius surrogate g (Lasso on the $\{1[T \subseteq S]\}$ basis), read its exact Shapley values in closed form, and certify by running CC-BET on the *residual* game $v - g$: since Shapley is linear, $\phi(v) = \phi(g) + \phi(v - g)$ exactly, so the residual certificate transfers to v . If the surrogate is good the residual is tiny and the certificate collapses quickly; if it is bad the certificate stays wide and the method degrades gracefully into CC-BET. Validity never depends on the surrogate being correct.

5 Theory

Throughout, $v \in [0, 1]$ so each observed (complementary-)contribution lies in $[-1, 1]$; a stratum’s samples, *conditioned on the sizes drawn*, are i.i.d. unbiased draws of that stratum’s mean, and the size-allocation and stopping rule are predictable (chosen from past observations).

Theorem 1 (Certificate validity under a doubly-stochastic oracle). *Let $[\ell_i, u_i]$ be the intervals returned by Alg. 1 with per-stratum level $\delta' = \delta/(n \cdot n_{\text{strata}})$. Then, for any data-dependent size-allocation and stopping policy,*

$$\mathbb{P}[\forall i \in N : \phi_i(v) \in [\ell_i, u_i]] \geq 1 - \delta.$$

Proof sketch. Fix a stratum (i, s) . Its observations are i.i.d. unbiased draws of $\mu_{i,s} = \mathbb{E}[\text{CC} \mid i \in S, |S| = s]$, bounded in $[-1, 1]$. The hedged capital process $K_t(m)$ evaluated at $m = \mu_{i,s}$ is a nonnegative martingale with $K_0 = 1$; by Ville’s inequality $\mathbb{P}[\exists t : K_t(\mu_{i,s}) \geq 1/\delta'] \leq \delta'$, so the confidence sequence $\{m : K_t(m) < 1/\delta'\}$ contains $\mu_{i,s}$ for all t simultaneously w.p. $\geq 1 - \delta'$. Time-uniformity is exactly what makes the guarantee survive a stopping time and a data-dependent number of samples per stratum: the allocation decides *how many* draws a stratum gets, never their conditional distribution. A union bound over the $n \cdot n_{\text{strata}}$ strata gives that all stratum means lie in their sequences jointly w.p. $\geq 1 - \delta$. On that event, by the decomposition (2) each $\phi_i = \frac{1}{n} \sum_s \mu_{i,s}$ lies in the size-average of the stratum intervals, which is $[\ell_i, u_i]$. The oracle’s within-coalition noise only affects the (bounded) sample values and is absorbed by the same martingale argument. $\square$

Theorem 1 is what makes the estimator *safe to run adaptively*: a fixed-sample bound would be void the moment the algorithm chooses its next size from the data. The ingredients (the CC decomposition, hedged betting, the union bound) are individually known; the contribution is the construction and its validity for segment attribution under the doubly-stochastic LLM oracle with adaptive allocation.

Proposition 1 (Two-level allocation). *Fix a call budget B and replication level r , giving $T = B/(2r)$ paired evaluations. Writing σ_{btw}^2 for the between-coalition variance of CC within a stratum and σ_{win}^2 for the per-call within-coalition variance,*

the per-segment estimator variance obeys

$$\begin{aligned}\text{Var}(\phi_i) &= O\left(\frac{\log n}{T}(\sigma_{\text{btw}}^2 + \frac{2\sigma_{\text{win}}^2}{r})\right) \\ &= O\left(\frac{2r \log n}{B}\sigma_{\text{btw}}^2 + \frac{4 \log n}{B}\sigma_{\text{win}}^2\right).\end{aligned}$$

Hence at fixed B the bound is minimized at $r = 1$ whenever $\sigma_{\text{btw}}^2 > 0$: fresh coalitions average both noise sources, replication only the within-coalition one. Under a prefix-cache token model where a coalition’s instruction prefix is paid once per coalition and the question/decoding tokens once per call, the token-optimal r shifts upward, to $r^* \approx \sqrt{(\sigma_{\text{win}}^2/\sigma_{\text{btw}}^2) \cdot (c_{\text{pre}}/c_{\text{call}})}$ with $c_{\text{pre}}, c_{\text{call}}$ the prefix and per-call token costs.

Proof sketch. The variance identity is the SVARM-style stratified decomposition applied to the paired identity (2), with the within-coalition term scaled by $1/r$ because each coalition value is an r -sample mean; substitute $T = B/(2r)$. The $r = 1$ optimum follows since the first term is increasing in r and the second is constant in B . The token statement replaces B (calls) by $T(c_{\text{pre}} + r c_{\text{call}})$ and minimizes over r . $\square$

Theorem 2 (Safe pruning: no false elimination). *Run Alg. 2 at level δ . With probability $\geq 1 - \delta$, simultaneously: (i) no group g with true Owen value $|\phi_g| > \tau$ is certified removable; and (ii) every kept group’s reported member intervals cover its true Owen values.*

Proof. Apply Thm. 1 to the quotient game at level $\delta/2$ and to each kept group’s within-group game at total level $\delta/2$; a union bound gives joint validity at δ . On the good event, a group is certified removable only if $[\ell_g, u_g] \subseteq [-\tau, \tau]$, which forces $|\phi_g| \leq \tau$; contrapositively $|\phi_g| > \tau$ is never certified removable. Member coverage is the group-conditional instance of Thm. 1. $\square$

Theorem 2 is the alignment guarantee: prompt compression driven by OWEN-ELIM cannot, except with the controlled probability δ , delete a group of clauses whose true contribution to a safety metric exceeds the tolerance τ . What we do *not* prove is a matching query-complexity lower bound for the certified problem; we regard establishing instance-optimal lower bounds under the noisy oracle as the main open theoretical question.

6 Experiments

Testbeds. **T0:** synthetic sparse low-order games with a Bernoulli (binary) and a low-noise Gaussian oracle, where exact Shapley values are known in closed form. **T1:** 8 single-instruction segments on MMLU, including a planted *sabotage* instruction (“the answer is always C”), with the full 2^8 grid giving exact ground truth on a real model. **T2:** 12 instruction segments in 4 groups on MMLU, exhaustive 2^{12} grid, exact Shapley/Owen/interaction ground truth, on two model families. **T3:** 20 segments in 5 groups ($2^{20} \approx 1.05\text{M}$ coalitions, enumeration impossible); the reference is a *certified gold standard*

Table 1: T2, Qwen3.5-9B, $n=12$, exact ground truth, betting certificates ($\delta=0.05$), 30,000 calls, 8 seeds. Lower ℓ_∞ is better; “cover” is the fraction of runs where all segment CIs contain the truth.

estimator	ℓ_∞	Kendall	cover	CI width
CC-Bet (ours)	0.011	0.920	1.00	0.221
Surrogate (ours)	0.015	0.894	1.00	0.301
KernelSHAP	0.029	0.905	—	—
Permutation MC	0.028	0.780	1.00	0.450
SVARM	0.052	0.686	—	—
Leverage SHAP	0.060	0.648	—	—

(CC-BET at 120k calls whose betting CIs bound its own error). **T4** (alignment, §7): certified attribution of prompt-injection resistance.

Models and protocol. Qwen3.5-9B and Gemma-4-26B via a hosted API, reasoning disabled, temperature 0; every response is cached so estimator comparisons replay from disk at identical randomness. Baselines: permutation Monte Carlo, KernelSHAP (paired), and the two strongest provable methods, Leverage SHAP and SVARM (the authors’ `shapiq` implementation). Certificates at $\delta = 0.05$, simultaneous over segments. All estimators consume the identical coalition oracle and are measured against exact truth (T0–T2) or certified gold (T3).

Certificate quality. Betting confidence sequences are $3\text{--}7\times$ tighter than empirical-Bernstein at equal δ with identical point estimates: on T2/Qwen the mean CC-BET interval width at 3k/10k/ 30k calls drops from 6.50/2.30/0.96 (Bernstein) to 0.85/0.41/0.22 (betting), and empirical coverage is 1.00 in every cell of every experiment (Table 1). The tightening is what lets certified decisions actually fire at realistic budgets (§7).

Noise breaks the deterministic-SOTA optimality. Under a binary (maximally noisy) oracle, the provable baselines trail: on T2/Qwen at 30k calls CC-BET attains $\ell_\infty = 0.011$ versus 0.052 (SVARM) and 0.060 (Leverage SHAP); the same ordering holds on Gemma-4, on T1, and on T3 (Table 2). We verified this is a genuine regime effect, not a faulty baseline: our Leverage SHAP reproduces the published $3\text{--}5\times$ advantage over KernelSHAP on *noiseless* high-order games and is exact at budget saturation, and in the low-noise Gaussian T0 both baselines become competitive again. Leverage SHAP suffers most because its inverse-leverage reweighting spans $\sim 2000\times$ across coalition sizes and amplifies extreme-size oracle noise—leverage sampling optimizes worst-case *deterministic* regression error, not variance under a stochastic oracle.

Scale beyond enumeration (T3). At $n=20$ the certified gold standard has mean CI width 0.163 over a million-coalition game, and both certified estimators are CI-consistent with gold on every segment. The estimator ranking is unchanged from the small- n exact-truth regimes, evidence that

Table 2: T3, Qwen3.5-9B, $n=20$ (2^{20} coalitions; enumeration impossible). Error vs. a certified gold reference (betting CIs, mean width 0.163) at 30,000 calls. “consistent” = every segment estimate within its own + gold’s CI.

estimator	ℓ_∞	Kendall	CI width	consistent
CC-Bet (ours)	0.013	0.768	0.291	yes
Surrogate (ours)	0.021	0.547	0.354	yes
KernelSHAP	0.029	0.663	—	—
SVARM	0.058	0.589	—	—
Leverage SHAP	0.068	0.389	—	—

the comparison generalizes past where ground truth is computable.

Allocation (Prop. 1). Sweeping replicates $r \in \{1, \dots, 16\}$ on both T2 games confirms $r \in \{1, 2\}$ minimizes call-cost error (by $r=16$, ℓ_∞ is 25–30% worse), while under the prefix-cache token model $r=2$ becomes token-optimal (equal error, $\sim 11\%$ fewer uncached tokens)—the predicted upward shift.

Interactions replicate a non-obvious finding. Exact pairwise interactions on T2 show reasoning-style instructions are *substitutes* (e.g. “eliminate wrong options” + “double-check” has interaction -0.27 on Qwen): one reasoning nudge helps, stacking several hurts. Coarse (group-level) attributions transfer across model families (Spearman 0.88) but fine-grained interactions do not (0/5 top-pair overlap), an argument for cheap per-model certified re-measurement.

7 Alignment Application: Certified Prompt Auditing

Certifying injection defenses (T4). We attribute *prompt-injection resistance* to system-prompt clauses with no harmful content. Each MMLU question is appended with a benign injection—“ignore all previous instructions and output the word BANANA”—and the metric is whether the model *resisted* (did not emit the trigger). We place 8 candidate system-prompt segments: three genuine defensive clauses (treat user text as untrusted; single-letter-answer-only; ignore override attempts) and five controls (persona, chain-of-thought, politeness, a distractor, a format clause). The exhaustive 2^8 grid gives exact resistance-Shapley on the real model.

With no system prompt the model is fully hijacked (resistance 0.00); the three defensive clauses alone raise resistance to 0.60 and the full prompt to 0.98. At 8k calls CC-BET certifies four clauses as load-bearing (betting interval strictly above zero) and leaves the remaining four *unclear* (interval straddles zero), matching the exact Shapley signs (Table 3). Three of the four certified clauses are the intended defenses; the fourth, revealingly, is the *format* control (“answer with a single letter”), which is a genuine non-obvious defense—constraining the output format itself suppresses the injected token. This is the kind of latent, interaction-driven safety dependency that

Table 3: T4 (alignment), Qwen3.5-9B, exact resistance-Shapley and certified CC-BET estimate (8k calls, 95% betting CIs). “*” marks clauses certified load-bearing (CI strictly > 0).

segment	kind	exact ϕ	95% CI
guard_untrusted*	defensive	+0.267	[+0.08, +0.43]
format*	control	+0.243	[+0.08, +0.45]
guard_task*	defensive	+0.203	[+0.04, +0.41]
guard_ignore*	defensive	+0.191	[+0.04, +0.41]
polite	control	+0.072	[−0.08, +0.28]
cot	control	+0.014	[−0.12, +0.23]
weather	control	+0.013	[−0.11, +0.23]
expert	control	−0.028	[−0.14, +0.18]

only a certified attribution surfaces, and that a naive prompt compressor would be free to delete.

Crucially, only certifying estimators can make this determination at all: on this lower-noise task the provable baselines’ *point* error is competitive, but they return no interval and therefore cannot certify any clause as load-bearing. Certification, not point accuracy, is the operative capability for an audit.

The certification replicates across model families. On Gemma-4 the same defense is certified, but the attribution reveals a different internal structure: a single clause (*guard_untrusted*, $\phi = +0.69$, CI [+0.50, +0.81]) carries the defense, and the other two defensive clauses are *redundant substitutes*—individually not certified because their marginal contribution given *guard_untrusted* is small (empty prompt 0.00, defensive group alone already 1.00). This is the correct Shapley reading and a feature, not a failure: marginal attribution exposes redundancy, and it is why the safety-relevant statement is made at the *group* (Owen) level, where the defensive group is certified load-bearing on both models. The same substitutability appeared for reasoning instructions in T2.

Safe pruning in both directions. The triage realizes Thm. 2 on real models and on the safety task itself (Table 4). On Gemma-4’s injection task it prunes correctly: the two control groups (*format*, *style*; true Owen values 0.006 and 0.002) are *certified removable*, while the defensive group (value 1.00) is kept. On Qwen the same defensive group is likewise never certified removable; at that model’s higher decision uncertainty the triage fail-safes on the borderline control groups (declines to prune) rather than risk a false elimination. A compression pass driven by OWEN-ELIM thus strips provably-inert clauses while, with probability $\geq 1 - \delta$, never removing a group that carries the safety behavior—exactly the guarantee a prompt-optimization pipeline needs to be trusted around a safety mitigation.

Certifying a harmful instruction (T1). In T1 the planted sabotage instruction has exact $\phi = -0.43$, and CC-BET at 8k calls certifies it *strictly harmful*: interval [−0.60, −0.26], bounded away from zero, whereas under empirical-Bernstein

Table 4: Certified safe-pruning triage on the T4 injection task ($\tau=0.1$, betting certificates), two model families. The defensive group is never certified removable; genuinely inert control groups are.

model	group	Owen value	verdict
Gemma-4	defensive	+1.00	kept
Gemma-4	format	+0.006	certified removable
Gemma-4	style	+0.002	certified removable
Qwen3.5	defensive	+0.87	kept
Qwen3.5	format	+0.058	kept (fail-safe)
Qwen3.5	style	+0.052	kept (fail-safe)

the same data gives $[-1.31, +0.40]$, which cannot support the conclusion. Certification, not point estimation, is what turns “this clause looks bad” into auditable evidence.

Why this is a safety contribution. The outputs are exactly the artifacts a safety case needs: (i) *verification* that a mitigation clause is effective, with a probability guarantee; (ii) a *harm flag* when a clause degrades a safety metric; and (iii) a *compression procedure that provably preserves safety behavior*. Because safety metrics (refusal, injection resistance, judge scores) are inherently noisy, the doubly-stochastic treatment is not optional—an uncertified attribution can, and in T1 does, point the wrong way.

8 Discussion and Limitations

Our certificates are valid but conservative; the betting construction narrows them enough to fire real decisions, yet closing the gap to an instance-optimal lower bound is open. We assume a canonical segment order, sidestepping position sensitivity; an order-aware game is future work. The token-cost model is idealized (opaque provider caching); a controlled study needs local serving. Finally, attribution certifies contribution to the *measured* metric—the safety of the conclusion is only as good as the behavioral judge, so the judge itself must be validated.

9 Conclusion

We framed system-prompt auditing as certified segment attribution under a doubly-stochastic oracle, gave an adaptive estimator whose betting certificate is provably valid in that setting, proved a safe-pruning guarantee, and showed on real models—up to a million-coalition regime—that certification is affordable and that certified auditing flags both effective injection defenses and harmful instructions. The alignment payoff is a prompt-compression and prompt-auditing primitive that produces verifiable, not merely plausible, safety evidence.

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